Involution Binary Relations *

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Abstract

This paper aims to investigate properties of strictly involution binary relations, which are generalizations of the classical notions of binary relations. We study the concepts of involution spanning subsets and give some examples related to different involution binary relations. Moreover, the involution independence with code is also considered in this paper.

Keywords: Antimorphic involution, Spanning sets, Primitive words, DNA Languages

1 Introduction

In this paper we study involution binary relations, which are generalizations of the classical notions of binary relations ([3],[4],[6]), motivated by the need of optimally encoding information as DNA strands for biomolecular computing purposes.

A DNA single strand is a linear chain made up of four different type of nucleotides, each consisting of a sugar-phosphate unit and a base (adenine, Cytosine, Guanine, or Thymine). A DNA single strand can be viewed as a string over the DNA alphabet of bases $\{\mathrm{A}, \mathrm{C}, \mathrm{G}, \mathrm{T}\}$. A DNA single strand has an orientation, with one end known as the 3' end, and the other known as the 5' end, based on their chemical properties. A word over the DNA alphabet represents a DNA single strand in its 5' to 3' orientation. An essential biochemical property of DNA single strands is that of Watson-Crick complementarity, where A can bind to T and C can bind to G by weak hydrogen bonds. Two Watson-Crick complementary DNA single strands of opposite orientation can bind to each other to form a DNA double strand.

One of the problems encountered when encoding information as DNA single strands is that the Watson-Crick complementarity often results in information-encoding DNA single strands either folding onto themselves to form intra-molecular structures, or interacting with each other to form inter-molecular structures. While these so-called secondary structures often have a significant role in determining the biochemical functions of real-life nucleic acids, in DNA computing they are often seen as a disadvantage. This is because it is very likely that the secondary structure formation of DNA strands will prevent then from interacting with other DNA strands in the expected, pre-programmed ways. Consequently, the property of a set of information-encoding strand to be free of unwanted intra-molecular and inter-molecular structures has been intensively studied from many different points of view [7].

2 Preliminaries

Let X be a finite alphabet and let X^* be the free monoid generated by X ([12],[13]). Any element of X^* is called a *word*. The length of a word w is denoted by $\lg(w)$. Any subset of X^* is called

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a *language*. Let $X^+ = X^* \setminus \{\lambda\}$, where λ is the empty word. An *involution* $\theta : X^* \to X^*$ of X^* is a mapping such that $\theta^2 = I$ where I is the identity mapping. It can extend to a morphic involution on X^* if for all $u, v \in X^*$, $\theta(uv) = \theta(u)\theta(v)$ or an antimorphic involution if $\theta(uv) = \theta(v)\theta(u)$. Note that $\theta(\lambda) = \lambda$.

We can look for repetitions of the form u^i for some word u and for some $i \geq 2$ in classical combinatorics on words. When dealing with DNA molecules (note that, their abstract representation as words), we have to take into account the fact that a word u encodes the same information as its complement $\theta(u)$, where θdenotes the Watson-Crick complementarity function, or its mathematical formalization as an arbitrary antimorphic involution. A θ-power of u is a word of the form $u_1u_2\cdots u_n$ for some $n \geq 1$, where $u_1 = u$ and for any $2 \leq i \leq n$, u_i is either u or $\theta(u)$. A θ-primitive word as strings which cannot be a θ-power of another word. The θ-primitive twin-root of a word w as the shortest word u such that w is a θ-power of u, denoted by $\sqrt[\theta]{w} = \{u, \theta(u)\}$. For instance, $w = AT = \theta(T)T = A\theta(A)$. This follows that $\sqrt[\theta]{AT} = \{A, T\} = \{A, \theta(A)\} = \{\theta(T), T\}$.

Let θ be either a morphic or an antimorphic involution on X^*. A involution binary relation ρ_θ on X^* is a subset of $\theta(X^*) \times X^*$. We define that the element $(\theta(u), v) \in \rho_\theta$ is denoted by the notation $u\rho_\theta v$ where ρ_θ is a involution binary relation. The element $(\theta(u), v) \notin \rho_\theta$ is denoted by the notation $u\widetilde{\rho_\theta}v$. We now recall some definitions introduced in ([8],[9]).

Definition 2.1 A involution binary relation ρ_θ is called a *strict involution binary relation* on X^* if for all $u, v \in X^*$,

(1) $u\rho_\theta u$ and $\lambda\rho_\theta u$;

(2) $u\rho_\theta v$ implies $\lg(\theta(u)) \leq \lg(v)$;

(3) $u\rho_\theta v$ and $\lg(\theta(u)) = \lg(v)$ imply $\theta(u) = v$.

There are some related involution binary relations defined as follow.

Definition 2.2 Let θ be either a morphic or an antimorphic involution on X^*.

(1) $w\rho_{\theta_\mathrm{u}}v$ if and only if $\theta(w) = v$ or $\lg(\theta(w)) < \lg(v)$;

(2) $w\rho_{\theta_\mathrm{p}}v$ if and only if $v \in \theta(w)X^*$;

(3) $w\rho_{\theta_\mathrm{s}}v$ if and only if $v \in X^*\theta(w)$;

(4) $w\rho_{\theta_\mathrm{i}}v$ if and only if $v \in X^*\theta(w)X^*$;

(5) $w\rho_{\theta_\mathrm{c}}v$ if and only if $v = \theta(x)w = wx$ for some $x \in X^*$;

(6) $w\rho_{\theta_\mathrm{d}}v$ if and only if $w\rho_\mathrm{p}v$ and $w\rho_{\theta_\mathrm{s}}v$, i.e., $v = x\theta(w) = wy$ for some $x, y \in X^*$;

(7) $w\rho_{\theta_\mathrm{b}}v$ if and only if $w\rho_{\theta_\mathrm{p}}v$ or $w\rho_{\theta_\mathrm{s}}v$.

A nonempty subset S of X^* is said to be *involution dependent* with respect to ρ_θ (shortly, *ρ_θ-dependent*) if there exist two distinct words u and v in S such that $u\rho_\theta v$. As an exception, let $\{\lambda\}$ be ρ_θ-dependent for any involution binary relation ρ_θ. A set $S \subseteq X^*$ is said to be *ρ_θ-independent* whenever S is not ρ_θ-dependent. The empty set and singleton sets except $\{\lambda\}$ all are involution independent with respect to any involution binary relation ρ_θ. The family of all ρ_θ-dependent (ρ_θ-independent) subsets of X^* is called a *ρ_θ-dependence (ρ_θ-independence)* in X^* and denoted by D_{ρ_θ} (I_{ρ_θ}).

A *ρ_θ-dependence relation*, $\sim_{\rho_\theta}$, is an involution binary relation defined on X^* such that $u \sim_{\rho_\theta} v$ if and only if $u\rho_\theta v$ or $v\rho_\theta u$.

Definition 2.3 Let θ be either a morphic or an antimorphic involution on X^* and let $S \in 2^{X^*} \setminus \{\emptyset, \{\lambda\}\}$. The *$\rho_\theta$-span* of S is defined by $\langle S\rangle_{\rho_\theta} = \cup_{v\in S\setminus\{\lambda\}}\{\theta(u) \in X^* | u \sim_{\rho_\theta} v\}$. Moreover, let $\langle\emptyset\rangle_{\rho_\theta} = \langle\{\lambda\}\rangle_{\rho_\theta} = \{\lambda\}$.

Proposition 2.1 Let θ be either a morphic or an antimorphic involution on X^* and let ρ_θ be an involution binary relation on X^*. Let $\{S_i\}_{i\in I}$, where I is an index set. Then

(1) $\langle\bigcup_{i\in I} S_i\rangle_{\rho_\theta} = \bigcup_{i\in I}\langle S_i\rangle_{\rho_\theta}$;

(2) $\langle\bigcap_{i\in I} S_i\rangle_{\rho_\theta} \subseteq \bigcap_{i\in I}\langle S_i\rangle_{\rho_\theta}$.

Proof.

(1) Let $\theta(u) \in \langle\bigcup_{i\in I} S_i\rangle_{\rho_\theta}$. There exists $v \in \bigcup_{i\in I} S_i$ such that $u \sim_{\rho_\theta} v$. Since $v \in \bigcup_{i\in I} S_i$, we have $v \in S_i$ for some i. This in conjunctive with $u \sim_{\rho_\theta} v$ yields that $\theta(u) \in \langle S_i\rangle_{\rho_\theta} \subseteq \bigcup_{i\in I}\langle S_i\rangle_{\rho_\theta}$. Thus $\langle\bigcup_{i\in I} S_i\rangle_{\rho_\theta} \subseteq \bigcup_{i\in I}\langle S_i\rangle_{\rho_\theta}$. Conversely, let $\theta(u) \in \bigcup_{i\in I}\langle S_i\rangle_{\rho_\theta}$. Then $\theta(u) \in \langle S_i\rangle_{\rho_\theta}$ for some i. There exists $v \in S_i$ such that $u \sim_{\rho_\theta} v$. Since $v \in S_i$, we have $v \in \bigcup_{i\in I} S_i$. This in conjunctive with $u \sim_{\rho_\theta} v$ yields that $\theta(u) \in \langle\bigcup_{i\in I} S_i\rangle_{\rho_\theta}$. Thus $\bigcup_{i\in I}\langle S_i\rangle_{\rho_\theta} \subseteq \langle\bigcup_{i\in I} S_i\rangle_{\rho_\theta}$. Therefore $\langle\bigcup_{i\in I} S_i\rangle_{\rho_\theta} = \bigcup_{i\in I}\langle S_i\rangle_{\rho_\theta}$.

(2) Let $\theta(u) \in \langle\bigcap_{i\in I} S_i\rangle_{\rho_\theta}$. There exists $v \in \bigcap_{i\in I} S_i$ such that $u \sim_{\rho_\theta} v$. Since $v \in \bigcap_{i\in I} S_i$,

we have $v \in S_i$ for all i. This implies that $\theta(u) \in \bigcap_{i\in I}\langle S_i\rangle_{\rho_\theta}$. Thus $\langle \bigcap_{i\in I} S_i\rangle_{\rho_\theta} \subseteq \bigcap_{i\in I}\langle S_i\rangle_{\rho_\theta}$. ∎

The statement $\bigcap_{i\in I}\langle S_i\rangle_{\rho_\theta} \subset \langle \bigcap_{i\in I} S_i\rangle_{\rho_\theta}$ is not true. We can see it from the following example.

Example 2.1 Let θ be a morphic involution on X^* and let ρ_θ be a involution binary relation on X^*. Let $X = \{A, T\}$, $S_1 = \{A, AT\}, S_2 = \{T, AT\}$, and $\rho_\theta = \{(A, AT), (T, AT), (AT, T), (AT, A)\}$. We have $\langle S_1\rangle_{\rho_\theta} = \{A, T, TA\}$, $\langle S_2\rangle_{\rho_\theta} = \{A, T, TA\}$, and $\langle S_1 \cap S_2\rangle_{\rho_\theta} = \langle\{AT\}\rangle_{\rho_\theta} = \{A, T\}$. Note that $TA \in \langle S_1\rangle_{\rho_\theta} \cap \langle S_2\rangle_{\rho_\theta}$ and $TA \notin \langle S_1 \cap S_2\rangle_{\rho_\theta}$.

Proposition 2.2 Let θ be either a morphic or an antimorphic involution on X^* and let ρ_θ be an involution binary relation on X^*. Let $S \subseteq X^*$ and let $\{\rho_{\theta_i}\}_{i\in I}$ a family of involution binary relations, where I is an index set. Then

(1) $\langle S\rangle_{\cup_{i\in I}\rho_{\theta_i}} = \bigcup_{i\in I}\langle S\rangle_{\rho_{\theta_i}}$;

(2) $\langle S\rangle_{\cap_{i\in I}\rho_{\theta_i}} \subseteq \bigcap_{i\in I}\langle S\rangle_{\rho_{\theta_i}}$.

Proof.

(1) Let $\theta(u) \in \langle S\rangle_{\cup_{i\in I}\rho_{\theta_i}}$. There exists $v \in S$ such that $(\theta(u), v) \in \rho_{\theta_i}$ for some i. It follows that $\theta(u) \in \bigcup_{i\in I}\langle S\rangle_{\rho_{\theta_i}}$. Thus $\langle S\rangle_{\cup_{i\in I}\rho_{\theta_i}} \subseteq \bigcup_{i\in I}\langle S\rangle_{\rho_{\theta_i}}$. Conversely, let $\theta(u) \in \bigcup_{i\in I}\langle S\rangle_{\rho_{\theta_i}}$. For $v \in S$, there exists $\theta(u) \in X^*$ such that $(\theta(u), v) \in \rho_{\theta_i}$ for some i. It follows that $(\theta(u), v) \in \bigcup_{i\in I}\rho_{\theta_i}$; hence $\theta(u) \in \langle S\rangle_{\cup_{i\in I}\rho_{\theta_i}}$. We have $\bigcup_{i\in I}\langle S\rangle_{\rho_{\theta_i}} \subseteq \langle S\rangle_{\cup_{i\in I}\rho_{\theta_i}}$. Then the statement (1) is true.

(2) Let $\theta(u) \in \langle S\rangle_{\cap_{i\in I}\rho_{\theta_i}}$. There exists $v \in S$ such that $(\theta(u), v) \in \rho_{\theta_i}$ for all i. It follows that $\theta(u) \in \bigcap_{i\in I}\langle S\rangle_{\rho_{\theta_i}}$. Thus $\langle S\rangle_{\cap_{i\in I}\rho_{\theta_i}} \subseteq \bigcap_{i\in I}\langle S\rangle_{\rho_{\theta_i}}$. ∎

The following example is an observation of $\bigcap_{i\in I}\langle S\rangle_{\rho_{\theta_i}} \not\subset \langle S\rangle_{\cap_{i\in I}\rho_{\theta_i}}$.

Example 2.2 Let θ be an antimorphic involution on X^* and let ρ_θ be a involution binary relation on X^*. Let $\rho_{\theta_0} = \{(\theta(w), w)|w \in X^*\} \cup \{(\lambda, w)|w \in X^+\} \cup \{(\theta(w), \lambda)|w \in X^+\}$. Let $X = \{A, T\}$, $\rho_{\theta_1} = \rho_{\theta_0} \cup \{(A, AT)\}$, and $\rho_{\theta_2} = \rho_{\theta_0} \cup \{(A, TA)\}$. Then we have $\langle\{TA, AT\}\rangle_{\rho_{\theta_1}} = \langle\{TA, AT\}\rangle_{\rho_{\theta_2}} = \{\lambda, T, AT, TA\}$. Moreover, $T \notin \langle\{TA, AT\}\rangle_{\rho_{\theta_1}\cap\rho_{\theta_2}} = \{\lambda, AT, TA\}$.

Furthermore, let $S \in X^+$. The definitions of minimal involution spanning subset and maximal involution independent subset of S are considered in the following.

Definition 2.4 Let θ be either a morphic or an antimorphic involution on X^* and $F = \{\langle S\rangle_{\rho_\theta} | S \in 2^{X^*}\}$. A subset A of $B \in F$ is said to be a ρ_θ-*spanning* subset of B if $\langle A\rangle_{\rho_\theta} = B$. Moreover, if $\langle A \setminus \{x\}\rangle_{\rho_\theta} \subset B$ for each $x \in A$, then A is said to be a *minimal* ρ_θ*-spanning* subset of B.

Let $S \subseteq X^*$ and $\rho_{\theta_1} \subseteq \rho_{\theta_2}$ be two involution binary relations defined on X^*. Then we have $\langle S\rangle_{\rho_{\theta_1}} \subseteq \langle S\rangle_{\rho_{\theta_2}}$.

Example 2.3 Let θ be an antimorphic involution on X^*. Let $X = \{A, T\}$. We have $\langle\{AT, TAT\}\rangle_{\rho_{\theta_\mathrm{p}}} \subseteq \langle\{T, TA, A^3\}\rangle_{\rho_{\theta_\mathrm{p}}}$. But $T^2AT \in \langle\{AT, TAT\}\rangle_{\rho_{\theta_\mathrm{b}}} \setminus \langle\{T, TA, A^3\}\rangle_{\rho_{\theta_\mathrm{b}}}$ and $A^2 \in \langle\{T, TA, A^3\}\rangle_{\rho_{\theta_\mathrm{b}}} \setminus \langle\{AT, TAT\}\rangle_{\rho_{\theta_\mathrm{b}}}$. Therefore, $\langle S_1\rangle_{\rho_{\theta_1}} \subseteq \langle S_2\rangle_{\rho_{\theta_1}}$ does not imply $\langle S_1\rangle_{\rho_{\theta_2}} \subseteq \langle S_2\rangle_{\rho_{\theta_2}}$, where $S_1, S_2 \subset X^*$.

Example 2.4 Let θ be an antimorphic involution on X^*. Let $X = \{A, T\}$. We have $\langle\{A, AT, T^2\}\rangle_{\rho_{\theta_\mathrm{p}}} = \langle\{A, T^2\}\rangle_{\rho_{\theta_\mathrm{u}}} = X^*$. It follows that $\{A, AT, T^2\}$ is a minimal ρ_{θ_p}-spanning subset of X^* but it is not a minimal ρ_{θ_u}-spanning subset of X^*. Furthermore, $\langle\{A, TA\}\rangle_{\rho_{\theta_\mathrm{p}}} = X^* \setminus ATX^*$ and $\langle\{A, TA\}\rangle_{\rho_{\theta_\mathrm{u}}} = X^*$. It follows that $\{A, TA\}$ is a minimal ρ_{θ_u}-spanning set of X^* but it is not a minimal ρ_{θ_p}-spanning subset of X^*. Therefore, a minimal ρ_{θ_1}-spanning subset of X^* does not imply that it is a minimal ρ_{θ_2}-spanning subset of X^* and vice versa.

3 Involution spanning Subsets

Let D_{ρ_θ} be a dependence in X^* and $S \subseteq X^*$. A dependence D_{ρ_θ} is said to be *transitive* if $\langle\langle S\rangle_{\rho_\theta}\rangle_{\rho_\theta} = \langle S\rangle_{\rho_\theta}$ for every $S \subseteq X^*$ ([2]). It is known that if D_{ρ_θ} is transitive, then all minimal ρ_θ-spanning subsets of $\langle S\rangle_{\rho_\theta}$ are ρ_θ-independent. Furthermore, let D_{ρ_θ} be not transitive. Is the ρ_θ-minimal spanning subset of $\langle S\rangle_{\rho_\theta}$ ρ_θ-independent ? Observe the following examples. We consider the subfamily ϕ'_θ of ϕ_θ, which is defined by $\phi'_\theta = \{\rho_\theta \in \phi_\theta |$ every minimal $\rho_\theta -$ spanning set with $\rho_\theta -$ independence$\}$.

Example 3.1 Let θ be an antimorphic involution on X^*. Let $X = \{A, T\}$.

(1) We have $\langle\{A, TA\}\rangle_{\rho_{\theta_\mathrm{u}}} = X^*, \langle\{A\}\rangle_{\rho_{\theta_\mathrm{u}}} = X^* \setminus \{A\}$ and $\langle\{TA\}\rangle_{\rho_{\theta_\mathrm{u}}} = X^* \setminus \{A^2, AT, T^2\}$. It follows that $\{A, TA\}$ is a minimal ρ_{θ_u}-spanning subset of X^*. Moreover, $\{A, TA\}$ is ρ_{θ_u}-dependent. Hence $\rho_{\theta_\mathrm{u}} \notin \phi'_\theta$.

(2) We have $\langle\{A, AT, T^2\}\rangle_{\rho_{\theta_p}} = X^*$. Moreover, $T^2 \in \langle\{A\}\rangle_{\rho_{\theta_p}} \setminus \langle\{AT, T^2\}\rangle_{\rho_{\theta_p}}, AT \in \langle\{AT\}\rangle_{\rho_{\theta_p}} \setminus \langle\{A, T^2\}\rangle_{\rho_{\theta_p}}$, and $A^2 \in \langle\{T^2\}\rangle_{\rho_{\theta_p}} \setminus \langle\{A, AT\}\rangle_{\rho_{\theta_p}}$. Similar, we have $\langle\{A, T\}\rangle_{\rho_{\theta_p}} = X^*$. It follows that $\{A, AT, T^2\}$ and $\{A, T\}$ are minimal ρ_{θ_p}-spanning subsets of X^*. These sets have not the same cardinality. However, $\{A, AT, T^2\}$ and $\{A, T\}$ are ρ_{θ_p}-dependent. Hence $\rho_{\theta_p} \notin \phi'_\theta$.

(3) We have $\langle\{A^2, T^2, AT, TA\}\rangle_{\rho_{\theta_p}} = X^*$. Moreover, the set $\{A^2, T^2, AT, TA\}$ is ρ_{θ_p}-independent.

Proposition 3.1 Let θ be either a morphic or an antimorphic involution on X^*. Let $\rho_\theta \in \phi_\theta$ and $w, v \in X^*$ with $\lg(w) \le \lg(v)$. If $\langle\{w\}\rangle_{\rho_\theta} \subseteq \langle\{v\}\rangle_{\rho_\theta}$ or $\langle\{v\}\rangle_{\rho_\theta} \subseteq \langle\{w\}\rangle_{\rho_\theta}$, then we have $w\rho_\theta v$.

Proof. Without loss of generality, we assume that $\langle\{w\}\rangle_{\rho_\theta} \subseteq \langle\{v\}\rangle_{\rho_\theta}$. There exists $x \in X^*$ such that $x \in \langle\{w\}\rangle_{\rho_\theta}$; hence $x \sim_{\rho_\theta} w$. It follows that $w\rho_\theta x$. Since $\langle\{w\}\rangle_{\rho_\theta} \subseteq \langle\{v\}\rangle_{\rho_\theta}$, $x \in \langle\{v\}\rangle_\rho$; hence $x \sim_{\rho_\theta} v$. We have $x\rho_\theta v$. From $w\rho_\theta x$ and $x\rho_\theta v$, these in conjunction with $\lg(w) \le \lg(v)$ yields that $w\rho_\theta v$. ∎

The converse of Proposition **3.1** is not true. Let ϕ_θ be the family of all involution binary relations on X^*.

Proposition 3.2 Let θ be either a morphic or an antimorphic involution on X^*. Then $\rho_\theta \in \phi'_\theta$ if and only if for $w, v \in X^*$ with $\lg(w) \le \lg(v)$, $w\rho_\theta v$ implies that $\langle\{w\}\rangle_{\rho_\theta} \subseteq \langle\{v\}\rangle_{\rho_\theta}$ or $\langle\{v\}\rangle_{\rho_\theta} \subseteq \langle\{w\}\rangle_{\rho_\theta}$.

Proof. ($\Rightarrow$) Let $\rho_\theta \in \phi'_\theta$ and let $w, v \in X^*$ with $w\rho_\theta v$ where $\lg(w) < \lg(v)$. Then $\{w, v\}$ is ρ_θ-dependent. Since $\rho_\theta \in \phi'_\theta$, we have $\{w, v\}$ in not a minimal ρ_θ-spanning set. It follows that $\langle\{w, v\}\rangle_{\rho_\theta} = \langle\{w\}\rangle_{\rho_\theta}$ or $\langle\{w, v\}\rangle_{\rho_\theta} = \langle\{v\}\rangle_{\rho_\theta}$. By Proposition 2.1(1), $\langle\{w, v\}\rangle_{\rho_\theta} = \langle\{w\}\rangle_{\rho_\theta} \cup \langle\{v\}\rangle_{\rho_\theta}$. Thus $\langle\{w\}\rangle_{\rho_\theta} \subseteq \langle\{v\}\rangle_{\rho_\theta}$ or $\langle\{v\}\rangle_{\rho_\theta} \subseteq \langle\{w\}\rangle_{\rho_\theta}$.

($\Leftarrow$) Let S be a minimal ρ_θ-spanning subset of X^*. Assume that there exist words $v, w \in S$ with $v \ne w$ such that $w\rho_\theta v$. Since $\langle\{w\}\rangle_{\rho_\theta} \subseteq \langle\{v\}\rangle_{\rho_\theta}$ or $\langle\{v\}\rangle_{\rho_\theta} \subseteq \langle\{w\}\rangle_{\rho_\theta}$, without loss of generality, we assume that $\langle\{v\}\rangle_{\rho_\theta} \subseteq \langle\{w\}\rangle_{\rho_\theta}$ holds. It follows that $\langle S \setminus \{v\}\rangle_{\rho_\theta} = \langle S\rangle_{\rho_\theta}$. This contradicts that S is a minimal ρ_θ-spanning subset of X^*. Thus $\rho_\theta \in \phi'_\theta$. ∎

4 Involution Independent Subsets

A non-empty subset S of X^* is called an involution independent set with respect to a involution binary relation ρ_θ defined on X^* or simply a ρ_θ-independent set if, for any $u, v \in S$, $u\rho_\theta v$ implies that $\theta(u) = v$.

Proposition 4.1 Let θ be either a morphic or an antimorphic involution on X^*. Let $\rho_{\theta_1}, \rho_{\theta_2}$ be two strict involution binary relations defined on X^*. Then $\rho_{\theta_1} \subseteq \rho_{\theta_2}$ on X^* if and only if $I_{\rho_{\theta_2}} \subseteq I_{\rho_{\theta_1}}$.

Proof. ($\Rightarrow$) Let $S \in I_{\rho_{\theta_2}}$. If S is a singleton set, then $S \in I_{\rho_{\theta_1}}$. Suppose that S contains at least two elements and let $u, v \in S$ such that $u\rho_{\theta_1}v$. Since $\rho_{\theta_1} \subseteq \rho_{\theta_2}$, we have $u\rho_{\theta_2}v$. This in conjunction with S being an involution independent set with respect to ρ_{θ_2} yields that $\theta(u) = v$. Thus $S \in I_{\rho_{\theta_1}}$; hence $I_{\rho_{\theta_2}} \subseteq I_{\rho_{\theta_1}}$.

($\Leftarrow$) Suppose that $u\rho_{\theta_1}v$ and $u\widetilde{\rho_{\theta_2}}v$ where $u, v \in X^+$. Then $\{\theta(u), v\} \in I_{\rho_{\theta_2}}$. Since $I_{\rho_{\theta_2}} \subseteq I_{\rho_{\theta_1}}$, it follows that $\{\theta(u), v\} \in I_{\rho_{\theta_1}}$. This implies that $u\widetilde{\rho_{\theta_1}}v$, a contradiction. Hence $u\rho_{\theta_2}v$. We have $\rho_{\theta_1} \subseteq \rho_{\theta_2}$ on X^*. ∎

Lemma 4.1([10]) Let X be an alphabet and let $x, y \in X^+$. Then $xy \ne yx$ if and only if $\{x, y\}$ is a code.

Definition 4.1 Let θ be either a morphic or an antimorphic involution on X^*. Let S' be a ρ_θ-independent subset of S. If $S' \cup \{x\}$ is ρ_θ-dependent for all $x \in S \setminus S'$, then S' is called a maximal ρ_θ-independent subset of S.

If S is ρ_θ-independent, then S is a ρ_θ-spanning subset. But a minimal ρ_θ-spanning subset does not imply that it is ρ_θ-independent. A maximal ρ_θ-independent subset S' of S with ρ_θ-spanning is called a ρ_θ*-basis* of S.

Example 4.1 Let θ be an antimorphic involution on X^*. Let $X = \{A, T\}$. The set $A^+ \cup A^+T^+ \setminus \{AT\}$ is a maximal ρ_{θ_p}-indepentent but not ρ_{θ_b}-indepentent subset of X^*. It follows that a maximal ρ_{θ_1}-independent subset of X^* does not imply that it is a maximal ρ_{θ_2}-independent subset of X^*.

5 Conclusions

In this paper, we study some properties of the involution binary relations. Some well known definitions and used results are presented in the second section. Section 3 is dedicated to the study of involution spanning subsets. There are some interesting examples considered for investigating the characterstic of involution spanning subsets.

Finally we consider the involution independence concepts related to codes.

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